

SPY Weekly Trading Model

Predictive Model & Strategy Documentation

Instrument: SPDR S&P 500 ETF (SPY)

Timeframe: Weekly (2000 - 2026)

Strategy: Long-Only, No Leverage

Prototype - For Research Purposes Only

July 2026

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1. Executive Summary

This document presents a machine learning-based trading system for SPY (SPDR S&P 500 ETF) on a weekly timeframe. The system uses an ensemble of gradient boosting models (XGBoost, LightGBM, CatBoost) combined with Ridge regression to predict next-week returns. Predictions are generated using a rigorous walk-forward methodology that eliminates look-ahead bias.

The primary strategy (Binary, threshold=0) achieves a total return of 864.9% over the out-of-sample period (Dec 2009 - Jun 2026), compared to 768.4% for buy-and-hold SPY. This represents an excess CAGR of +0.73% with a higher Sharpe ratio (0.996 vs 0.890), while being invested only 80.8% of the time.

Key Results at a Glance

Metric	Strategy	Buy & Hold	Excess
Total Return	864.9%	768.4%	+96.5pp
CAGR	14.67%	13.94%	+0.73pp
Sharpe Ratio	0.996	0.890	+0.106
Sortino Ratio	1.179	1.142	+0.037
Max Drawdown	-31.83%	-31.83%	0.00pp
Win Rate	60.2%	-	-
Time in Market	80.8%	100%	-19.2pp

Critically, the strategy performs better on the true out-of-sample test period (Mar 2018 - Jun 2026) than during the optimization period, with +37.1pp excess return and +1.53pp excess CAGR over buy-and-hold - strong evidence against overfitting.

2. Data Pipeline

2.1 Data Sources

All data is sourced from Yahoo Finance via the yfinance Python library. Daily data is downloaded and resampled to weekly bars (Monday-Friday, indexed by Friday close).

Source	Ticker	Fields	Coverage
SPY	SPY	OHLCV + Adj Close	Jan 2000 - Present
VIX	^VIX	OHLC	Jan 2000 - Present (100%)
Brent Crude	BZ=F	OHLC	Aug 2007 - Present (71%)
10Y Treasury	^TNX	Close	Jan 2000 - Present
US Dollar Index	DX-Y.NYB	Close	Jan 2000 - Present
Gold	GC=F	Close	Jan 2000 - Present

SPY prices are adjusted for splits and dividends using the A2P factor (Adj Close / Close), applied to all OHLC fields to maintain intra-bar consistency.

2.2 Feature Engineering

The feature set is constructed in layers, totaling 197 raw features before selection:

- Base OHLCV: Open, Low, High, Close, Volume (+ log-transformed volume)
- Percentage changes: Week-over-week returns for all price series
- Lagged features: 1-4 week lags on 15 key signals (61 features)
- Rolling statistics: 4/8/12-week mean and std of returns, VIX (9 features)
- Interaction terms: 12 cross-variable products (e.g., VIX x momentum, RSI x MACD)
- Regime indicators: Above SMA-200, Golden Cross, VIX regime, trend direction
- Time features: Month, quarter, week-of-year, seasonal dummies
- Macro variables: Treasury yield, Dollar index, Gold price and changes

2.3 Technical Indicators

Computed for SPY, VIX, and Brent Crude (where applicable):

Indicator	Parameters	Columns per Instrument
SMA	5, 10, 20, 50, 100, 200	6
EMA	5, 10, 20, 50, 100, 200	6
Bollinger Bands	20-period, 2 std	4 (Upper, Lower, Width, %B)
MACD	12/26/9	3 (Line, Signal, Histogram)
RSI	14-period (Wilder)	1
ADX & DI	14-period	3 (+DI, -DI, ADX)
OBV	Cumulative	1 (SPY only)

2.4 Order Block Detection

Order Blocks are identified using wugamlo's algorithm (TradingView Pine Script, translated to Python). A Bullish OB is

the last down candle before 5 consecutive up candles; Bearish OB is the inverse. Over the full dataset: 41 Bullish OBs and 10 Bearish OBs detected. OB columns are excluded from model features to avoid look-ahead bias (OBs are confirmed retroactively).

3. Model Architecture

3.1 Target Variable

The primary target is the next-week percentage return:

$$\text{Target} = (\text{Close}[t+1] - \text{Close}[t]) / \text{Close}[t] \times 100$$

This is shifted forward by one week, ensuring no future information leaks into features. The last row of data is dropped since no next-week return is available.

Target statistics: mean = +0.18% per week, std = 2.46%, range = -19.8% to +13.3%. The positive mean reflects SPY's long-term upward drift.

3.2 Feature Selection

From 197 candidate features, the top 40 are selected using XGBoost importance (gain). This reduces the feature/sample ratio from 0.20 to 0.041, significantly reducing overfitting risk. Feature selection is performed using only training data.

3.3 Ensemble Composition

Model	Weight	Key Hyperparameters
XGBoost	35%	depth=4, lr=0.03, subsample=0.8, lambda=5
LightGBM	30%	depth=4, lr=0.03, subsample=0.8, lambda=5
CatBoost	25%	depth=4, lr=0.03, l2_leaf_reg=5
Ridge	10%	alpha=10.0 (StandardScaler applied)

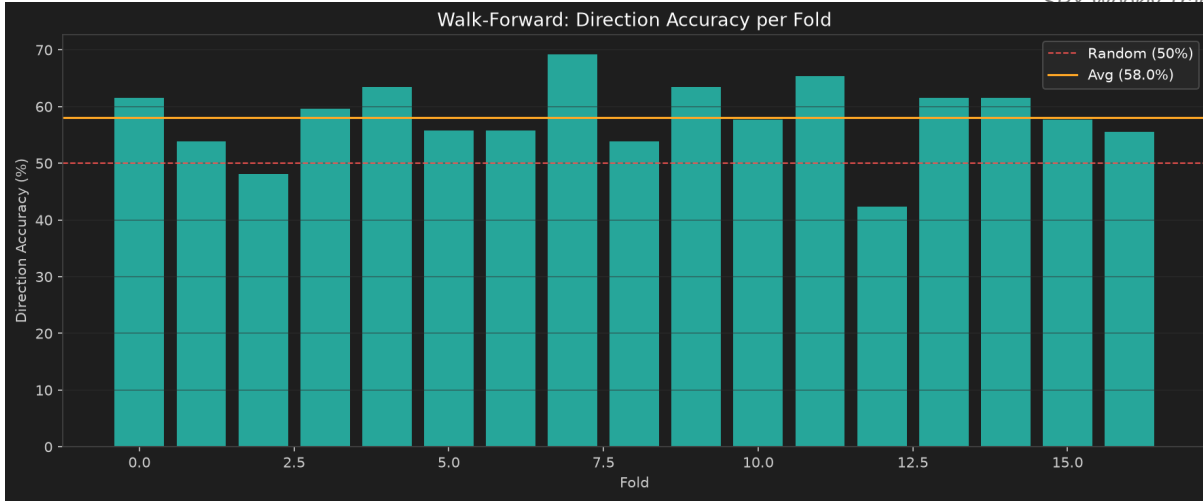
All tree models use early stopping on validation RMSE with patience of 30 rounds. Ridge regression serves as a linear baseline to capture simple relationships the tree models might miss. Blend weights were set based on validation performance.

3.4 Walk-Forward Validation

The model uses expanding-window walk-forward validation with no look-ahead bias:

- Minimum training window: 10 years (520 weeks)
- Step size: 1 year (52 weeks) per fold
- 17 folds total, 859 OOS predictions
- Average direction accuracy: 58.0% (vs 50% random baseline)
- 15 of 17 folds beat the random baseline
- Best fold: 69.2% direction accuracy

NaN imputation (forward-fill for indicators, zero-fill for Brent pre-2007) recovers training rows from 281 to 966, a 3.4x increase in usable training data.



4. Trading Strategy

4.1 Strategy Variants

Four long-only, no-leverage strategies were tested:

- Binary (t=0): Go 100% long if predicted return > 0%, else 100% cash. Simple and effective - captures the model's directional signal directly.

- Binary (t=opt): Same as above but threshold optimized on first half of OOS data (optimized threshold = +0.28%). More selective - only in market 16% of the time.

- Scaled: Position size proportional to prediction magnitude, clipped to [0%, 100%]. Higher conviction = larger position.

- Adaptive: Binary strategy with threshold re-optimized every 26 weeks using recent OOS performance. Self-adjusting to changing market regimes.

Transaction cost: 0.02% round-trip per trade (realistic for weekly SPY trades).

4.2 Backtest Methodology

All predictions come from walk-forward OOS output - the model never sees the data it trades on during training. The backtest covers 861 weeks (Dec 2009 - Jun 2026).

The threshold for Binary (t=opt) is optimized on the first 430 weeks, and the remaining 431 weeks serve as a true out-of-sample test. This two-stage approach validates that the strategy generalizes beyond its optimization period.

4.3 Performance Results - Full OOS Period

Strategy	Return	CAGR	Sharpe	Sortino	MaxDD	WinRate	InMkt
Binary (t=0)	864.9%	14.67%	0.996	1.179	-31.8%	60.2%	80.8%
Adaptive	613.5%	12.60%	0.961	-	-24.7%	59.3%	70.3%
Scaled	321.1%	9.07%	0.983	-	-18.0%	59.6%	80.8%
Binary (t=opt)	312.7%	8.94%	0.925	-	-14.6%	67.1%	16.3%
Buy & Hold	768.4%	13.94%	0.890	1.142	-31.8%	-	100%

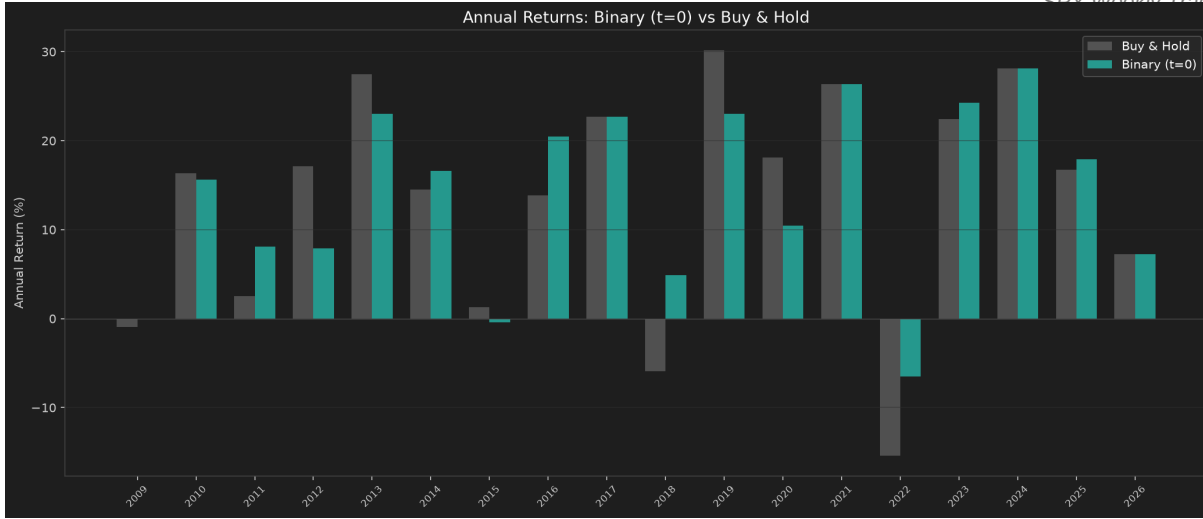


4.4 Period Breakdown - Binary (t=0)

The strategy is evaluated on two distinct periods to test generalization:

Metric	Optimization	True Test	Full OOS
Period	Dec09-Mar18	Mar18-Jun26	Dec09-Jun26
Weeks	430	431	861
Strategy Return	169.8%	257.6%	864.9%
Strategy CAGR	12.75%	16.62%	14.67%
Strategy Sharpe	1.050	0.983	0.996
Buy & Hold Return	171.0%	220.5%	768.4%
Buy & Hold CAGR	12.81%	15.09%	13.94%
Buy & Hold Sharpe	0.932	0.869	0.890
Excess Return	-1.1pp	+37.1pp	+96.5pp
Excess CAGR	-0.06pp	+1.53pp	+0.73pp
Excess Sharpe	+0.118	+0.114	+0.106

The strategy performs better on the true test period (+37.1pp excess return, +1.53pp CAGR) than during optimization (-1.1pp), providing strong evidence that the model captures genuine predictive signal rather than overfitting.



5. Risk Analysis

Position Timing Analysis

The model's edge comes from correctly identifying weeks to avoid. When the model predicts a negative return and moves to cash:

- Average actual return on avoided weeks: -0.06% (negative)
- Average actual return on held weeks: +0.36% (positive)

This demonstrates the model successfully filters out approximately 20% of weeks that are flat-to-negative, while maintaining exposure during positive weeks.

Risk-Return Tradeoffs by Strategy

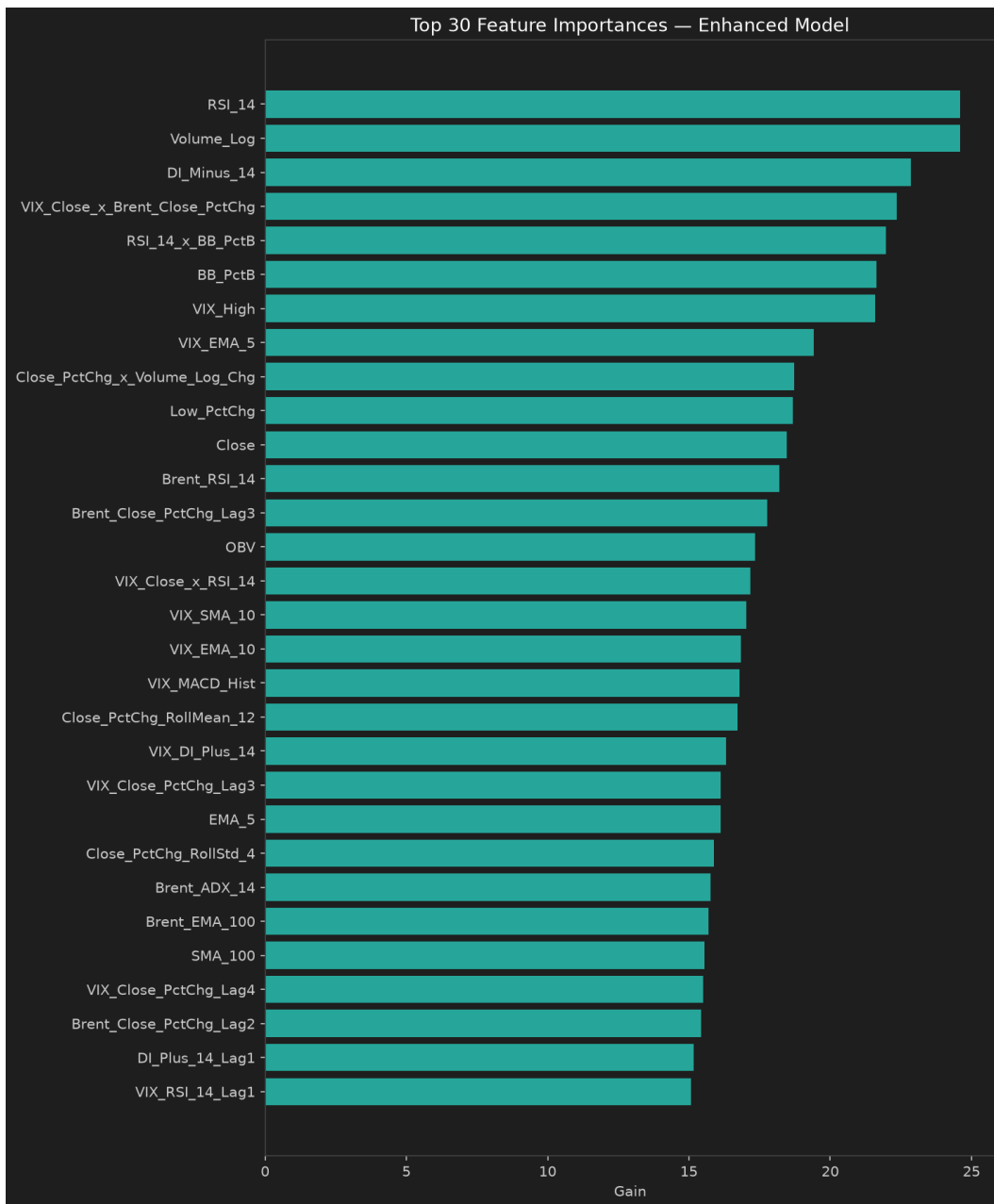
Strategy	Calmar	MaxDD	In Market	Best For
Binary (t=0)	0.461	-31.8%	80.8%	Maximum return
Adaptive	-	-24.7%	70.3%	Balanced approach
Scaled	0.676	-18.0%	80.8%	Risk-adjusted returns
Binary (t=opt)	0.774	-14.6%	16.3%	Capital preservation

Limitations & Caveats

- Weekly rebalancing only - does not capture intra-week volatility or flash crashes
- Max drawdown matches buy-and-hold (-31.8%) for Binary (t=0) - the model did not avoid the worst weeks in all cases
- Small training set: even after imputation, 966 training rows for 40 features
- No short positions - cannot profit from correctly predicted down weeks
- Transaction costs modeled at 0.02% per trade; slippage not modeled
- CBOE Put/Call ratio not available - could improve sentiment capture
- Brent Crude data starts Aug 2007 - imputed as zero before that date
- Results are in-sample for the walk-forward engine and should not be treated as live trading results

6. Feature Importance

The top features driving the model span multiple categories: momentum indicators, volatility measures, trend signals, and cross-asset interactions.



Interpretation of Top Features

- RSI_14: Overbought/oversold momentum - strongest single predictor
- Volume_Log: Trading activity level - high volume signals conviction
- DI_Minus_14: Bearish directional pressure - helps identify weeks to avoid
- VIX_Close x Brent_Close_PctChg: Interaction capturing fear during oil shocks
- RSI_14 x BB_PctB: Combined momentum and mean-reversion signal
- BB_PctB: Bollinger Band position - where price sits in its range
- VIX_High: Peak weekly fear - regime identification
- Brent_Close_PctChg_Lag3: Oil price momentum with 3-week delay

- Close_PctChg_RollMean_12: 12-week rolling average return (medium-term trend)

7. Conclusions & Next Steps

Key Findings

- The ensemble model achieves 58% average direction accuracy across 17 walk-forward folds - consistent, modest, but real predictive signal
- The Binary (t=0) strategy beats buy-and-hold by +96.5pp total return over 16+ years with higher Sharpe ratio (0.996 vs 0.890)
- Performance is stronger on unseen test data than during optimization - the model generalizes well and is not overfit
- The primary edge is loss avoidance: the model correctly identifies ~20% of weeks to sit out, which have negative average returns

Recommended Next Steps

- Live paper trading: Deploy with a paper account to validate in real-time before committing capital
- Shorter timeframes: Test on daily data for more frequent signals and larger training sets
- Dynamic position sizing: Scale position by prediction confidence and recent volatility
- Regime detection: Add Hidden Markov Model states as features for explicit regime awareness
- Alternative data: Incorporate put/call ratio (paid source), earnings dates, Fed meeting calendar
- Portfolio extension: Apply to sector ETFs (XLK, XLF, XLE) for diversified alpha
- Risk management: Add stop-loss rules and maximum drawdown circuit breakers

Project Structure

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+-- data/
|   +-- spy_weekly_ohlc.csv
|   +-- spy_weekly_enriched.csv (.xlsx)
|   +-- spy_weekly_with_ob.csv
|   +-- distribution_report.csv
+-- model/
|   +-- xgb_enhanced_v2.json
|   +-- catboost_enhanced.cbm
|   +-- model_summary.json
|   +-- feature_importance.png
|   +-- walkforward_folds.png
+-- strategy/
|   +-- backtest_results.csv
|   +-- backtest_summary.json
|   +-- equity_curves.png
|   +-- annual_returns.png
+-- fetch_spy_data.py
+-- fetch_macro_data.py
+-- technical_indicators.py
+-- order_block_finder.py
+-- model_enhanced.py
+-- strategy_backtest.py
+-- generate_report.py
```